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BENGALURU CAMPUS

Gen AI & LLMs Bootcamp

MeitY
Government of India

Bootcamp on Gen AI and LLMs: Concepts and Applications
1st April to 4th April 2026

GenAI for Images – Introduction to Diffusion models

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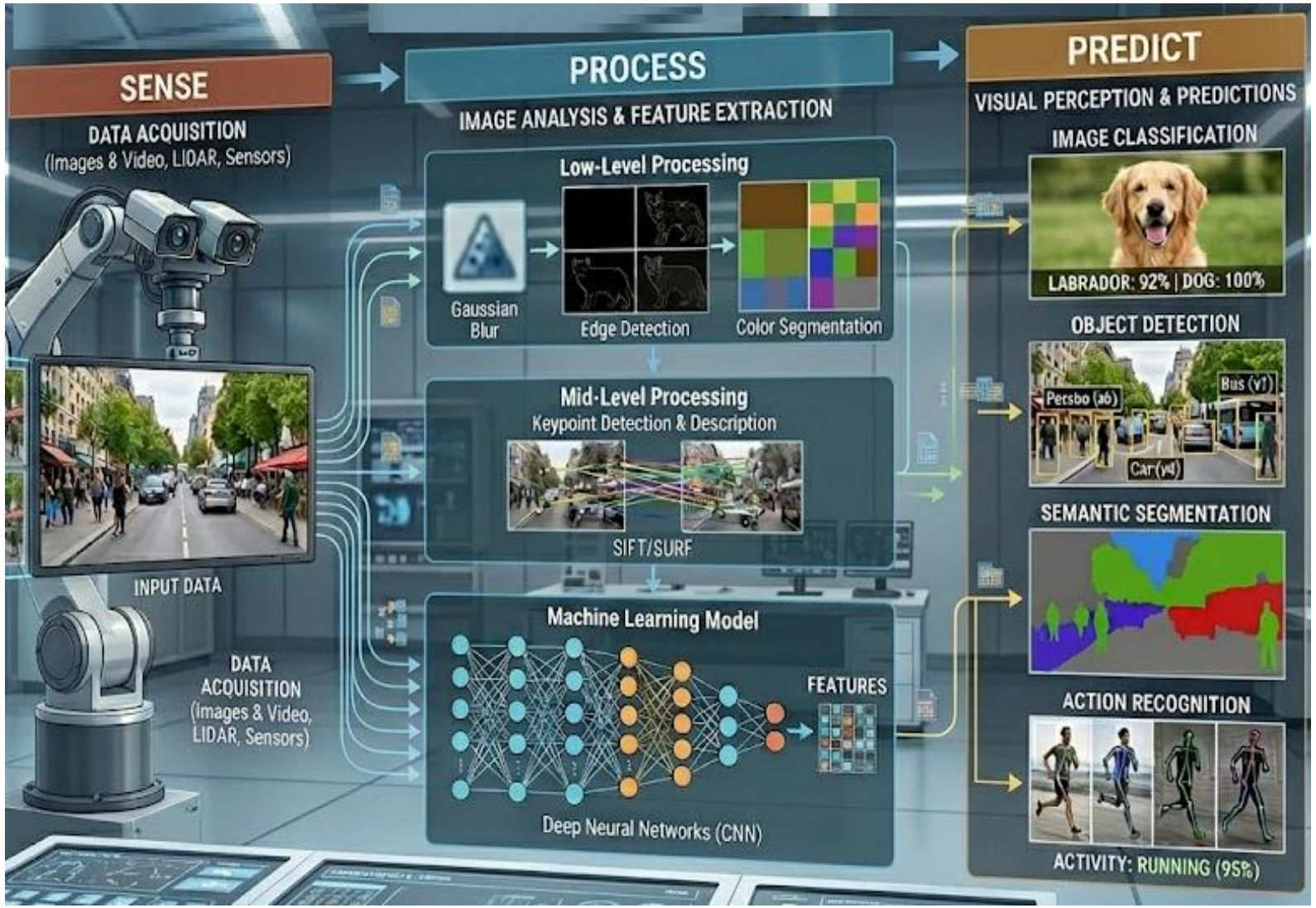
AI for image processing: Pre and Post Gen-AI era



➢ Pre-GenAI

- **Typical Tasks**
- *Image classification* : Is this a Dog or a Cat?
- *Object detection* : Identifying where is the object
- *Face recognition:* Whose face is present in the image
- *Medical diagnosis :* Segmenting the impacted skin regions.

- **Objective:**
- Learn P(y|x)
- Learn the probability of an answer *given* what we observe.

Earlier AI *looked at images*.

AI for image processing: Pre and Post Gen-AI era

➢ Post-GenAI

- **Typical Tasks**
- *Image to Image generation*
- *Text → Image generation:* Create an image of a Cat in a field
- *Image editing :* Adding effects to the image
- *Style transfer :* Applying filters, converting photos into toons
- *Synthetic* data creation

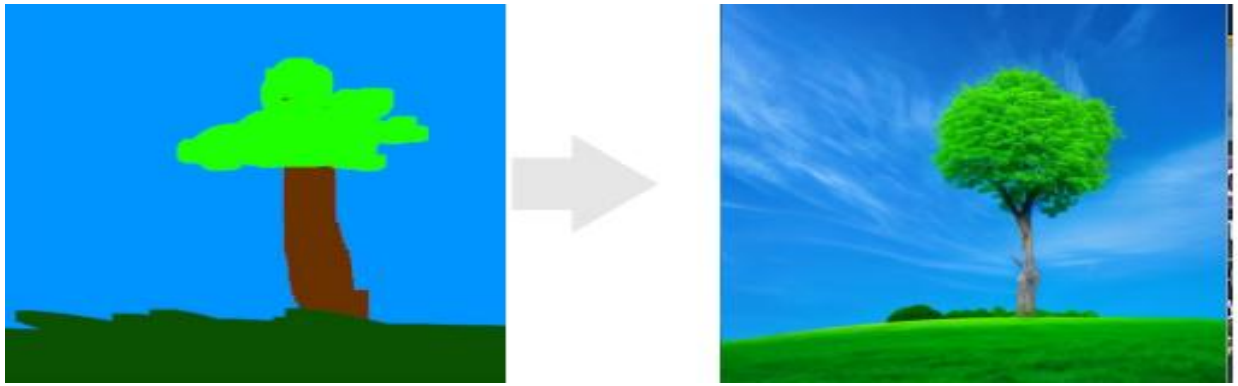
- **Objective:**
- Learn P(x)
- Learn the distribution of image itself

Modern AI *imagines images*.

AI has shifted from understanding the world → to creating new worlds.

Generative AI

AI has shifted from understanding the world → to creating new worlds.



Objective:

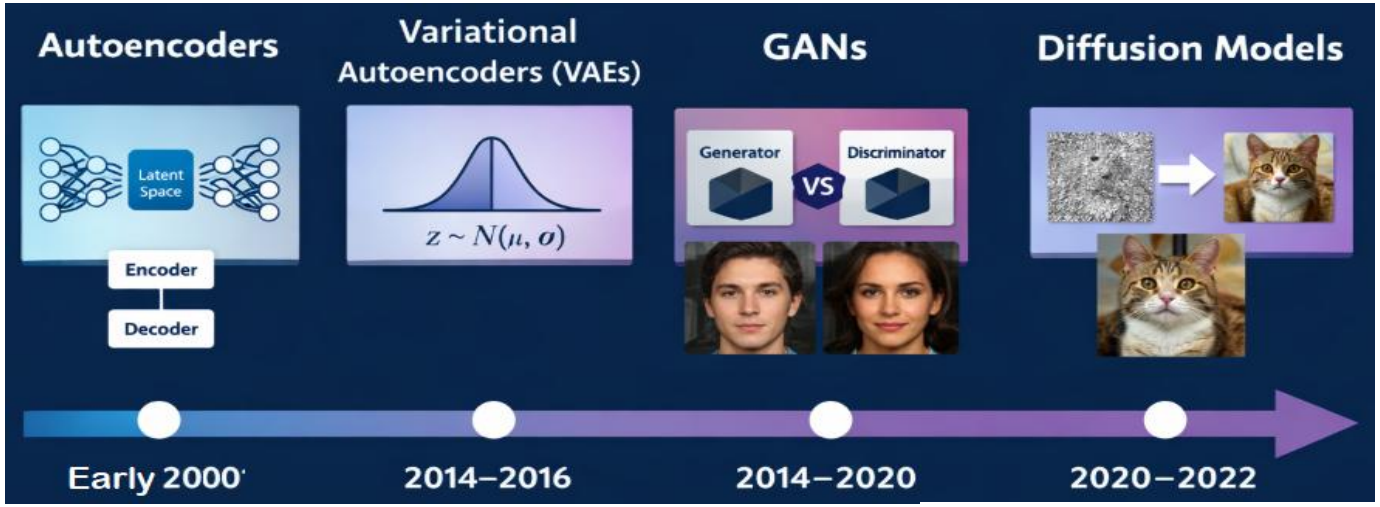
- 'How can I create an image that looks like it could exist?'
- Learns P(x) — the distribution of all possible images.
- By learning this distribution, the AI can generate completely new, realistic images.

- **Typical tasks**
- Creative Art & Design AI-assisted artwork, illustrations, digital painting
- Product & Fashion Design - Generate prototypes, concept sketches, patterns
- Gaming & Entertainment Imaginary worlds, characters, environments
- Medical & Scientific Visualization Enhance images, generate synthetic datasets
- Film & Advertising Concept art, storyboarding, ad visuals

Think of AI as a collaborative co-creator: it doesn't replace humans, it amplifies creativity and speeds up the creative process.

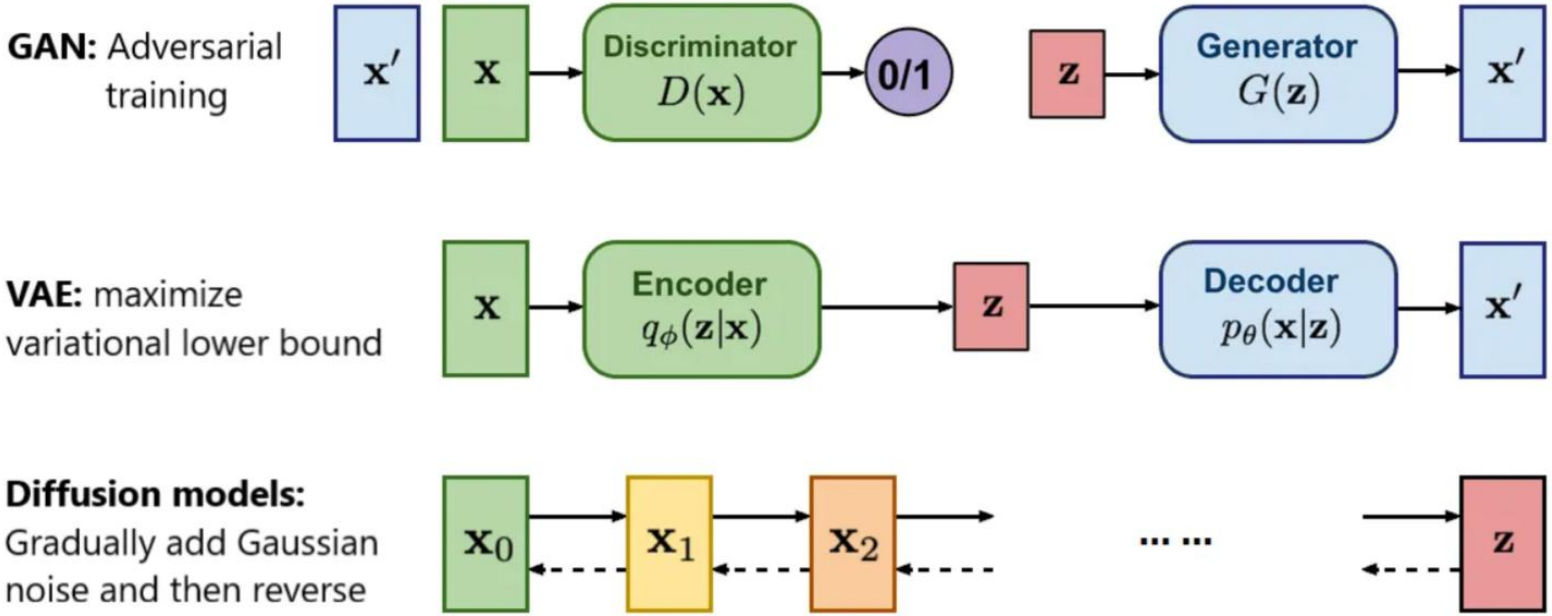
Imagine you could instantly create any image you want — what would you generate first?

Generative AI: Timeline

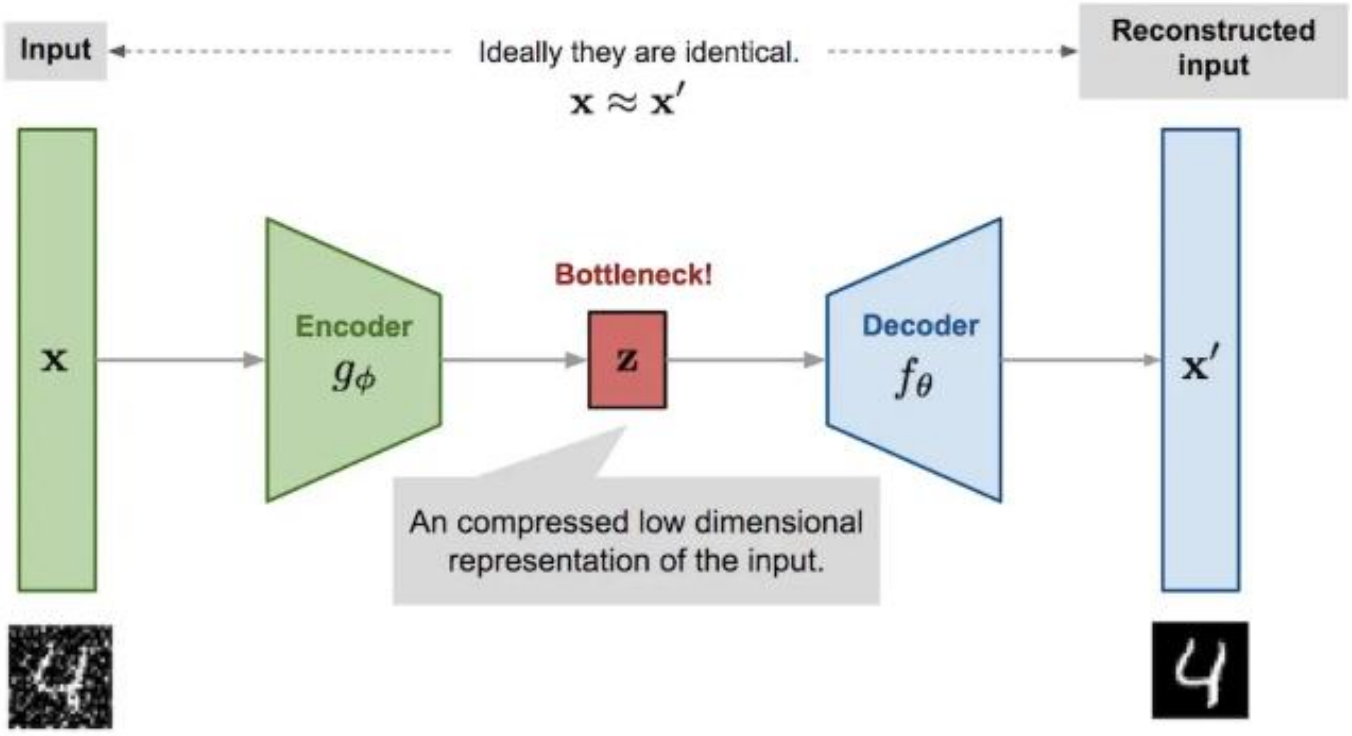


- Early 2000s: Autoencoders –typical encoder-decoder architectures
- 2013–2014: VAEs introduced probabilistic latent spaces

- 2014–2021: GANs revolutionized realistic image generation
- 2021–Present: Diffusion models dominate high-quality image creation



Autoencoders

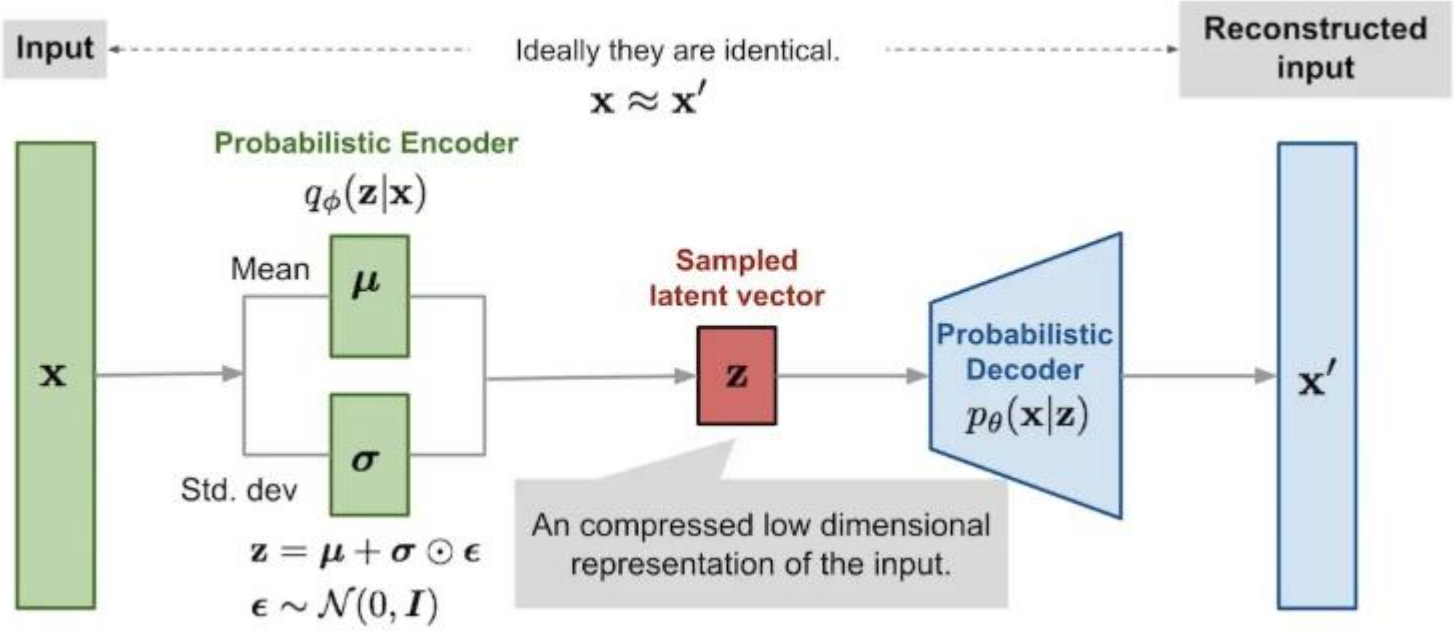


Autoencoder architecture with the bottleneck layer

Learning to Reconstruct

- Compresses input to a fixed **latent space**
- Learns **reconstruction** from latent representation.
- Goal: f(x) ≈ X
- Creates data from **the scope** of the training dataset
- **Usecases:** Denoising, anomaly detection, data compression
- **Limitation:** Outputs often blurry; cannot generate truly new images

Variational Autoencoders



Probabilistic latent space: μ, σ

Encoder generates a **distribution**

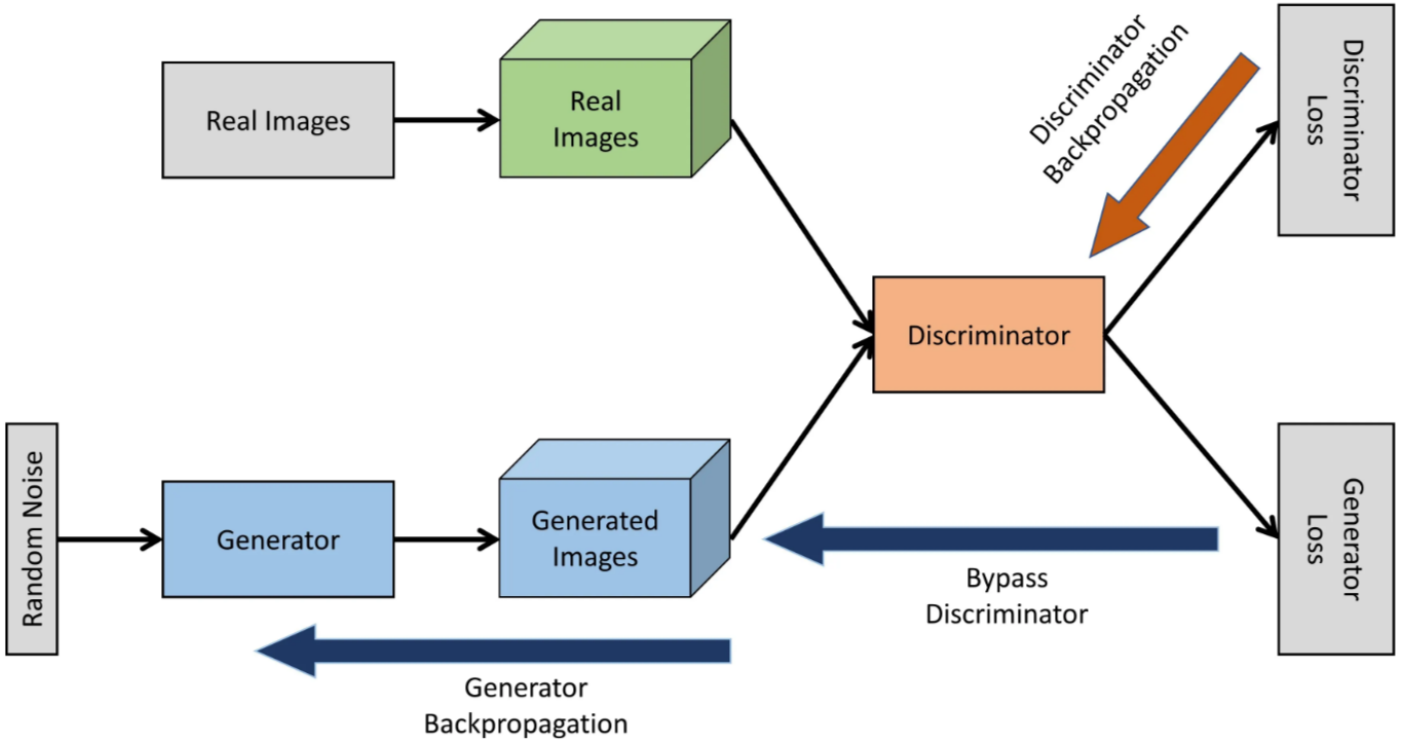
Reconstruction: Samples latent vector to generate new image

Sampling from this distribution allows VAEs to **generate new images**, not just reconstruct existing ones.

Usecases: Compression, Image generation, denoising

Limitation: Can generate new images, but still sometimes blurry

Generative Adversarial Networks



Adversarial Learning – Ian Goodfellow
Two networks : Generator vs Discriminator

Generator: creates fake images

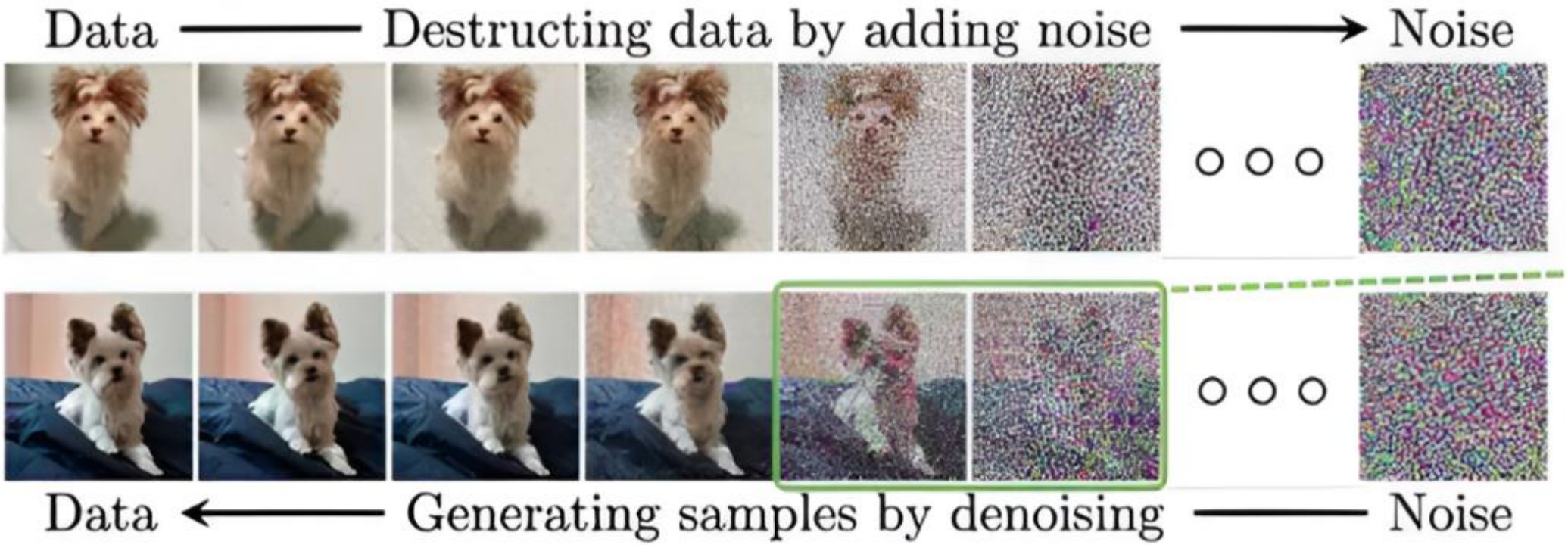
Discriminator: Evaluates and distinguishes real vs fake

Training: adversarial, both improve together . Discriminator and Generator compete.

Produces **highly realistic images**

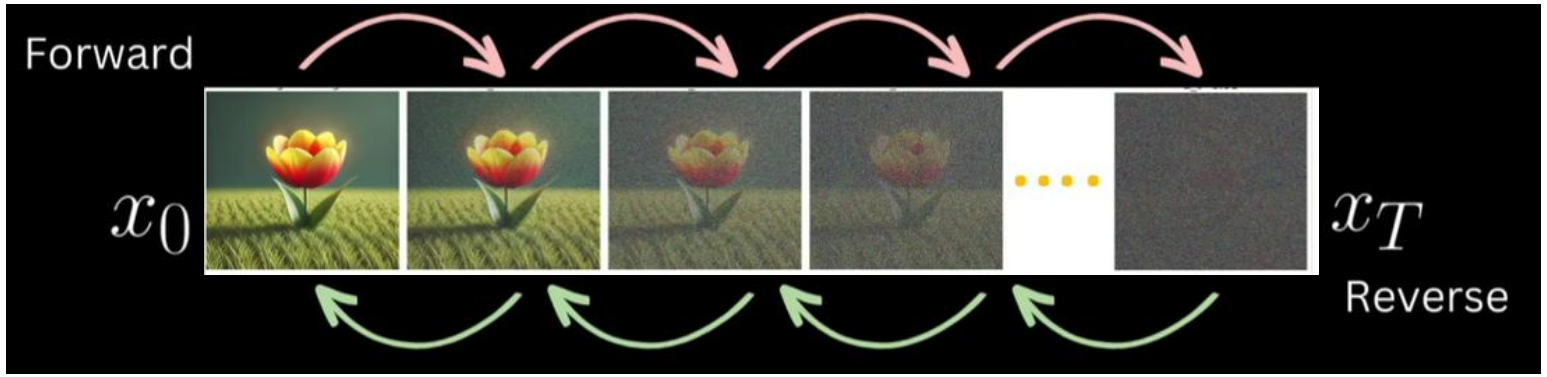
Limitation: However, training is unstable and sometimes the generator collapses, producing limited diversity."

Diffusion Models



- **Diffusion models** are inspired by the physical process of particle spread out due to random motion
- Operate through a two-phase process:
  - **Forward process:** Add noise gradually to data
  - **Reverse Process:** Denoises to generate new samples from pure noise.
- Models learn to recover original data distribution from the noisy distribution

Diffusion Models: Forward process

- **Forward process:** Gradually adds Gaussian noise to the given image through different timesteps
- 

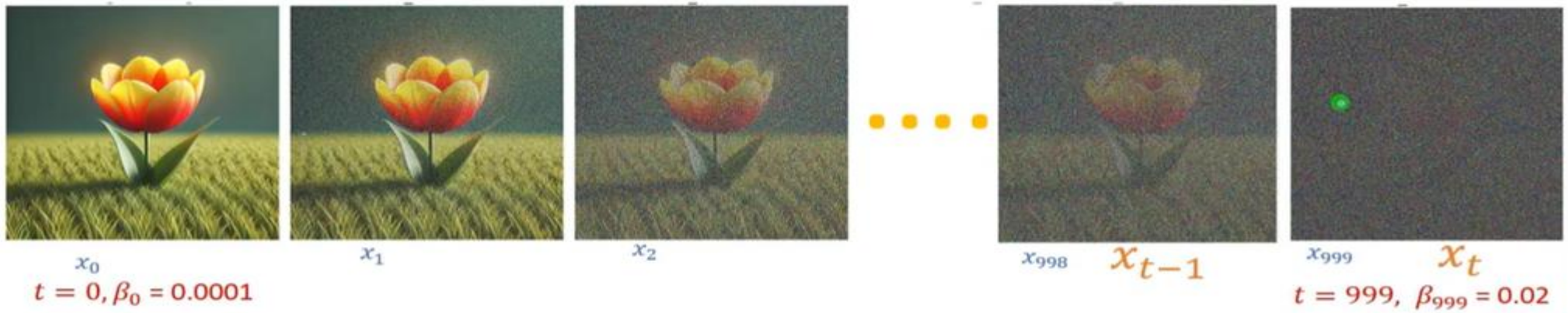
$$q(x_t|x_{t-1}) = N\left(x_t; \sqrt{1-\beta_t}.x_{t-1}, \beta_t.I\right)$$

- Using this, model tries to represent simpler distribution from complex distribution.

- What does Beta represent?
- How is it scheduled?
- Is the same amount of noise added at each step?

Diffusion Models: Forward process

- **What is Beta????**



- Calculating Beta.

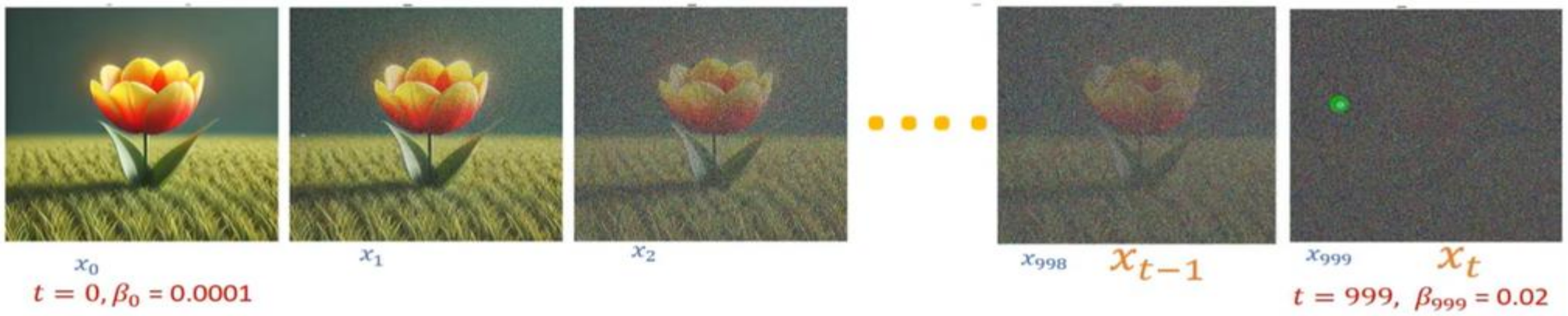
$$\beta_t = \beta_{start} + t \times \frac{\beta_{end} - \beta_{start}}{T}$$

- T=1000, β_{start} =0.0001, β_{end} =0.02.
- At t=0, β_0=0.0001.
- At t=999, β_{999} =0.02.

T=0, 0.0001
T=1, 0.0001199
T=2, 0.0001398
.
.
T=999, 0.02

Diffusion Models: Forward process

- **What is Beta????**



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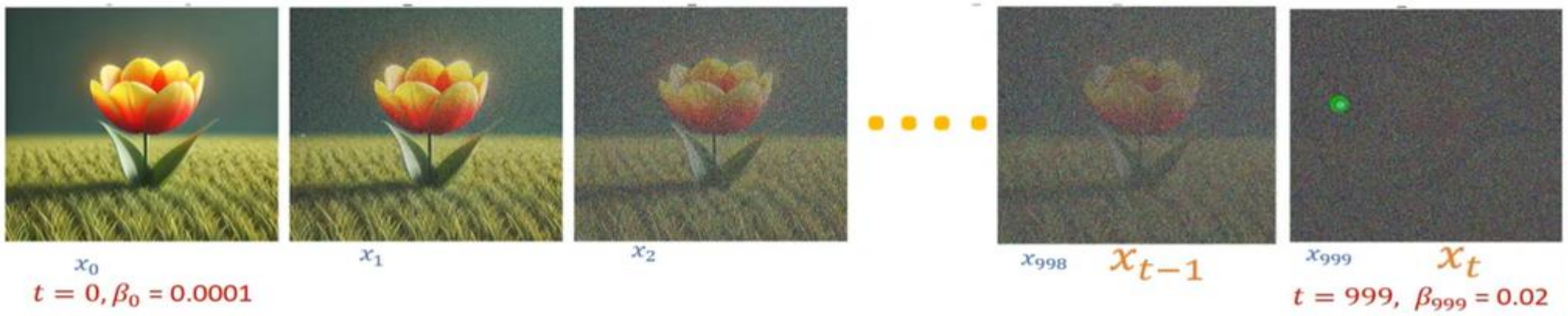
- What does Beta represent? ------- Amount of Noise
- How is it scheduled? ---------- Linearly
- Is the same amount of noise added at each step? ----------NO

Repeat the process for 1000 steps

Diffusion Models: Forward process

- **What is Beta????**



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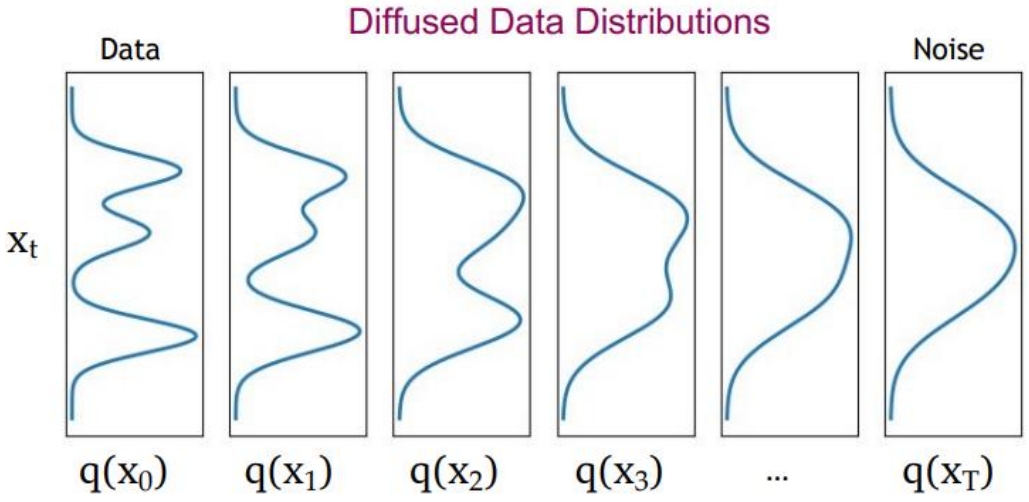


Diffusion Models: Forward process

$$q(x_t|x_{t-1}) = N(x_t, \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

$$\boldsymbol{\alpha}_t = 1 - \beta_t$$

$$x_t = \sqrt{1-\beta_t}.x_{t-1} + \sqrt{\beta_t}.\boldsymbol{\epsilon}$$

ϵ : $\epsilon \sim N(0,1)$ epsilon is random Gaussian noise.

$$= \sqrt{\alpha_t}.x_{t-1} + \sqrt{1-\alpha_t}.\boldsymbol{\epsilon}$$

$$= \sqrt{\alpha_t \boldsymbol{\alpha}_{t-1}}x_{t-2} + \sqrt{1-\alpha_t\boldsymbol{\alpha}_{t-1}}\boldsymbol{\epsilon}$$

$$= \sqrt{\alpha_t \boldsymbol{\alpha}_{t-1}\boldsymbol{\alpha}_{t-2}}x_{t-3} + \sqrt{1-\alpha_t\boldsymbol{\alpha}_{t-1}\boldsymbol{\alpha}_{t-2}}\boldsymbol{\epsilon}$$

$$\overline{\alpha_t} = \prod_{s=1}^{t} \alpha_s$$

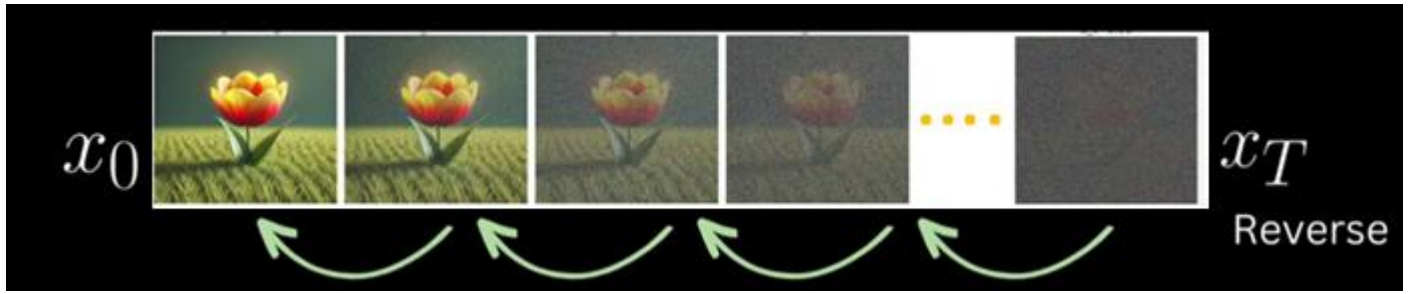
$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$$

$$q(x_t|x_0) = N(x_t, \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$$

Is this actually repeated for 1000 steps???
Computationally inefficient

Diffusion Models: Reverse process

Reverse Process: Denoises the input image gradually generate new image samples from pure noise.



$$p_\theta(x_{t-1}|x_t) = N\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right)$$

- But we **cannot compute reverse** distribution.
- Approximation through the neural networks.
- Model **learns to predict** Noise in the image sample

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)\right\|^2\right]$$

Why predicting noise????

Diffusion Models: Reverse process

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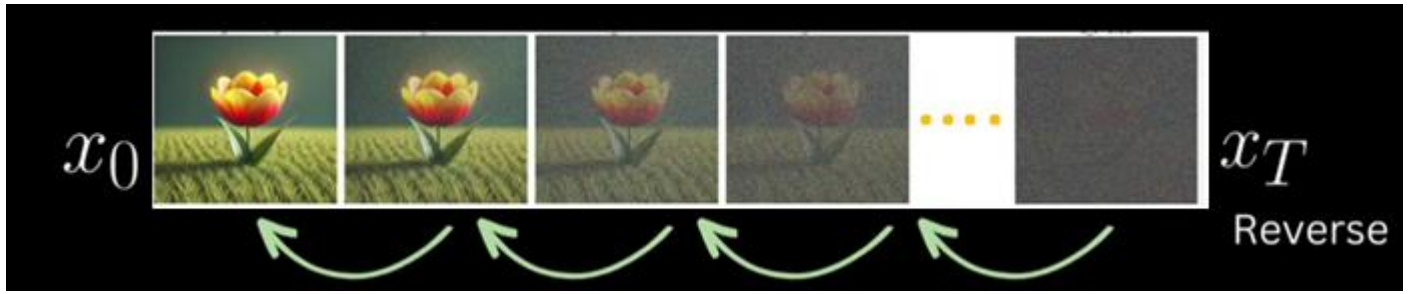

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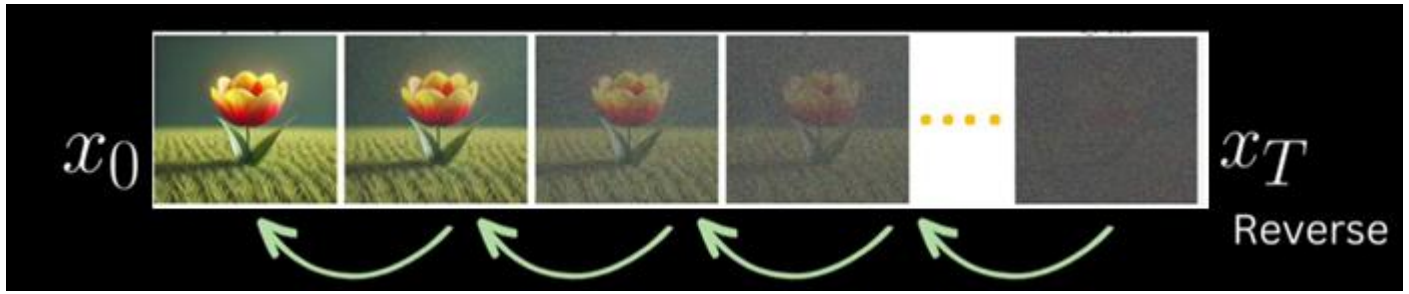
Why predicting noise????
Easier to model, stable, controllable generation

Diffusion Models: Training process



Algorithm 1 Training

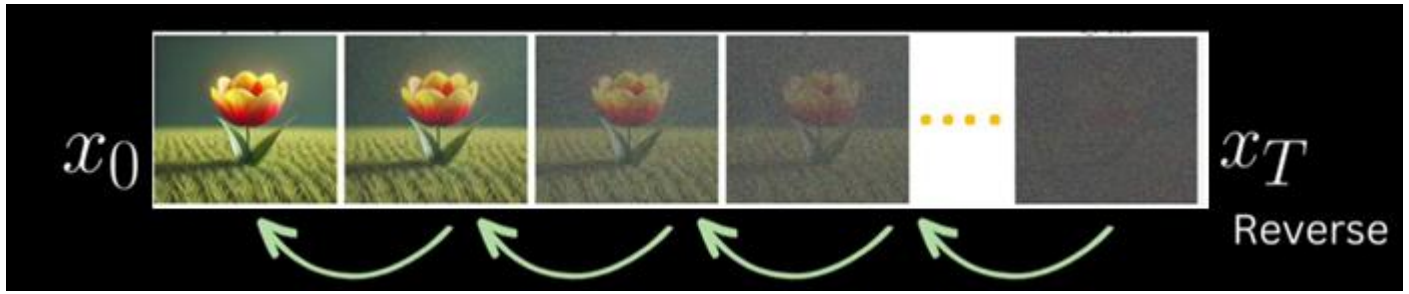
1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on

$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$

6: **until** converged

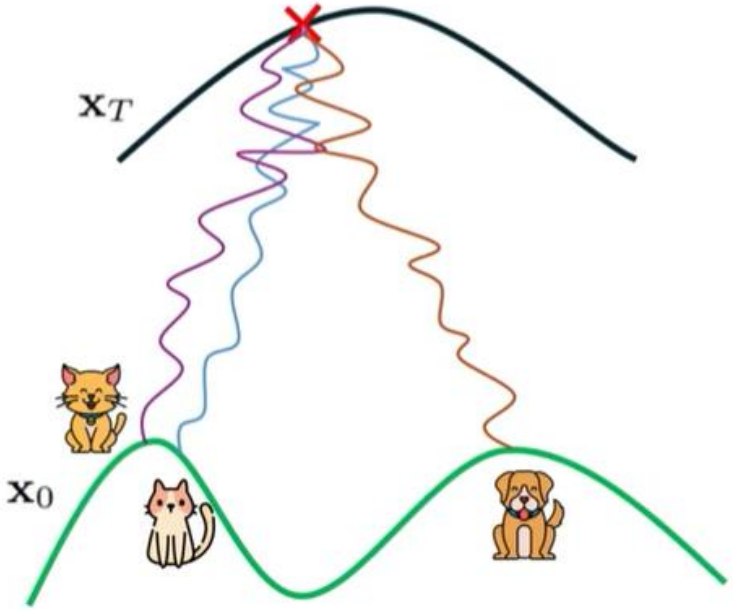
Diffusion Models: Training process

.



Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\quad \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

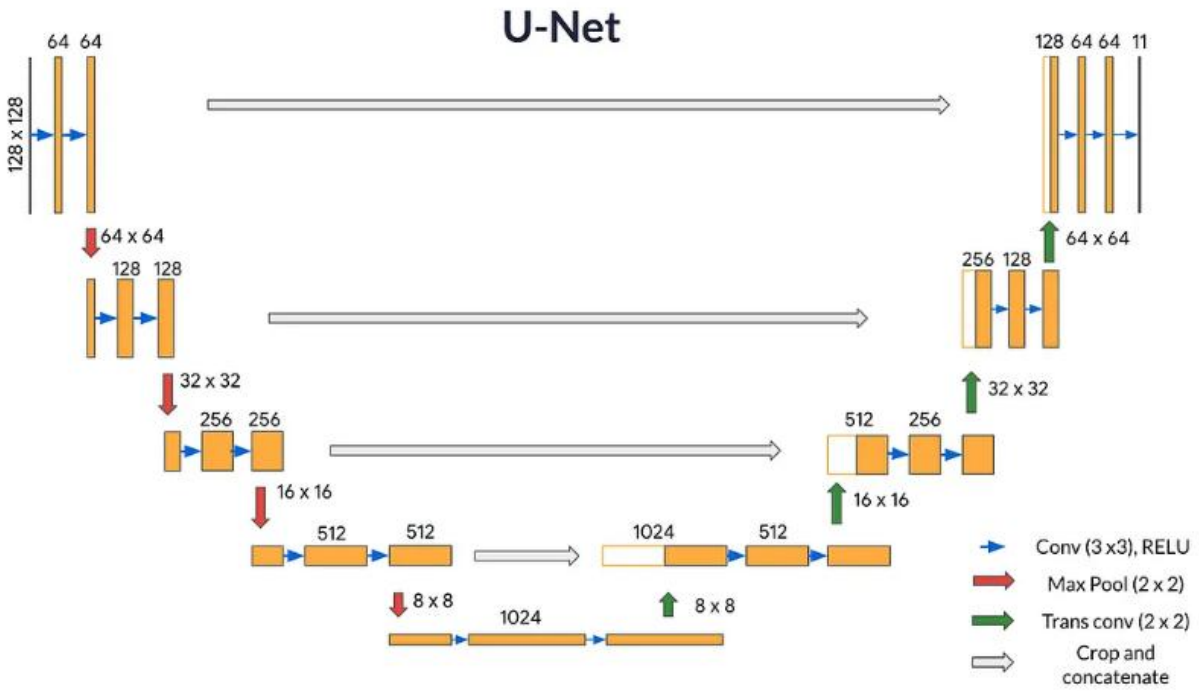


Diffusion Models: Relation to UNets

- U-Net architecture learns the denoising process.

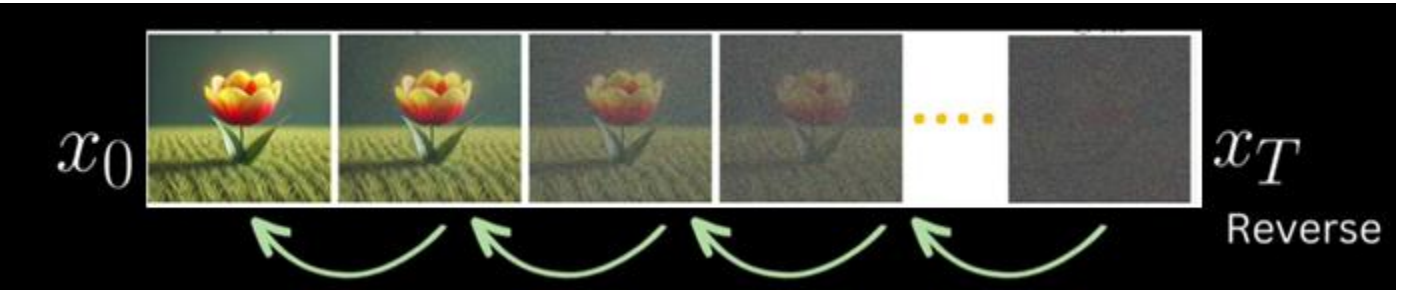
- Input: Noisy image and time stamp

- Output: predicted Noise



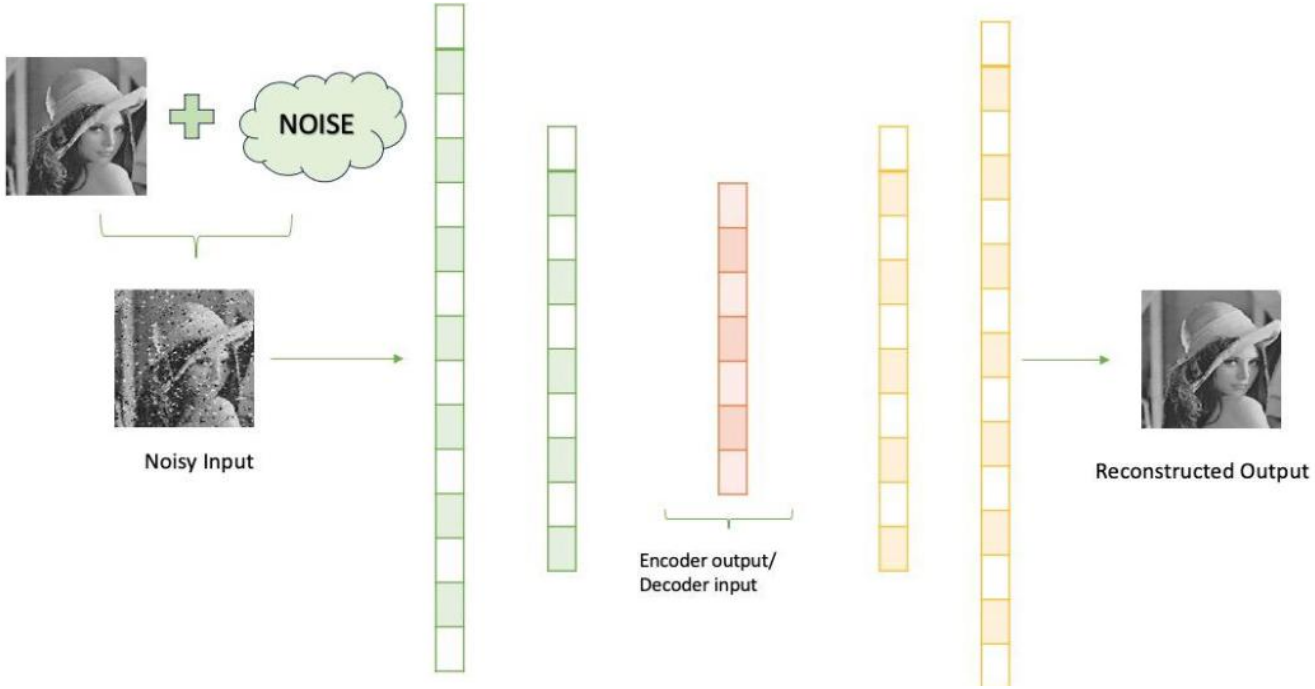
The U-Net architecture

Diffusion Models and Autoencoders



Are they similar to Denoising Autoencoders??????

- Each step is tiny denoiser
- Diffusion is sequence of denoising autoencoders
- Learns multi-scale noise removal



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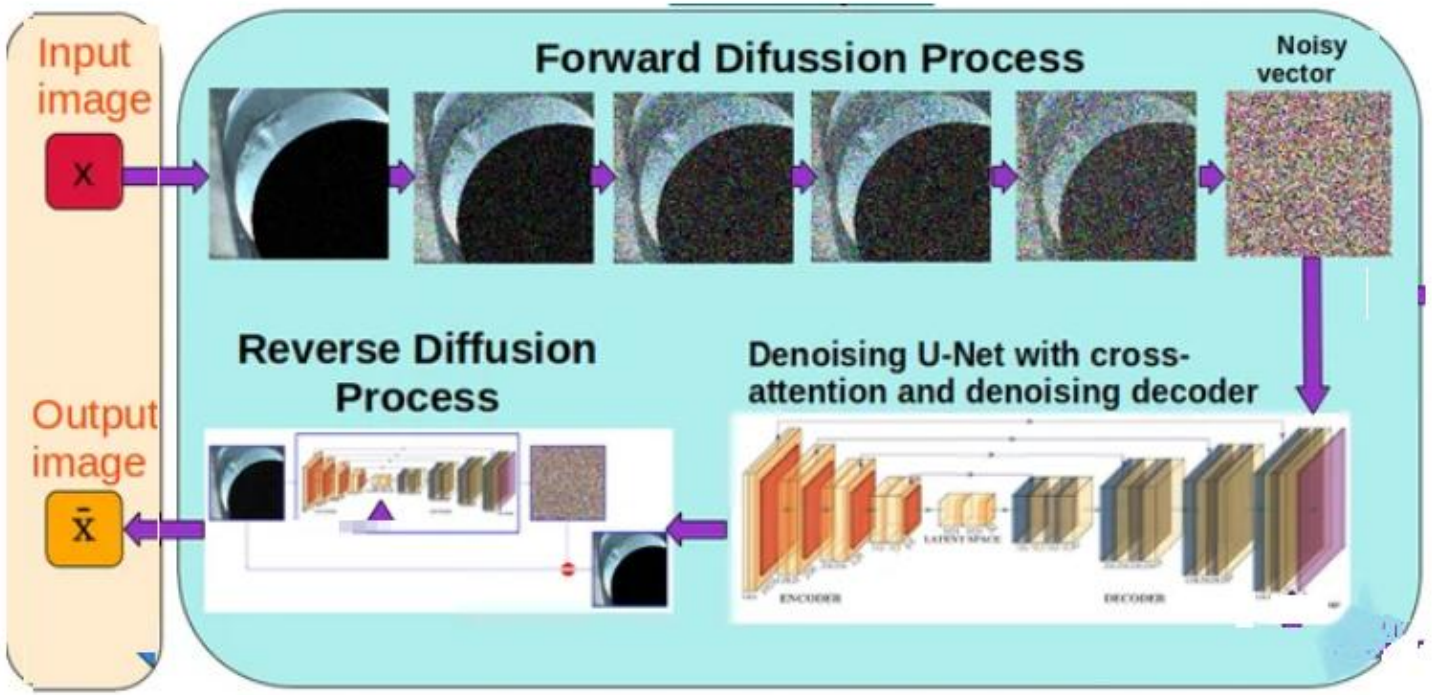
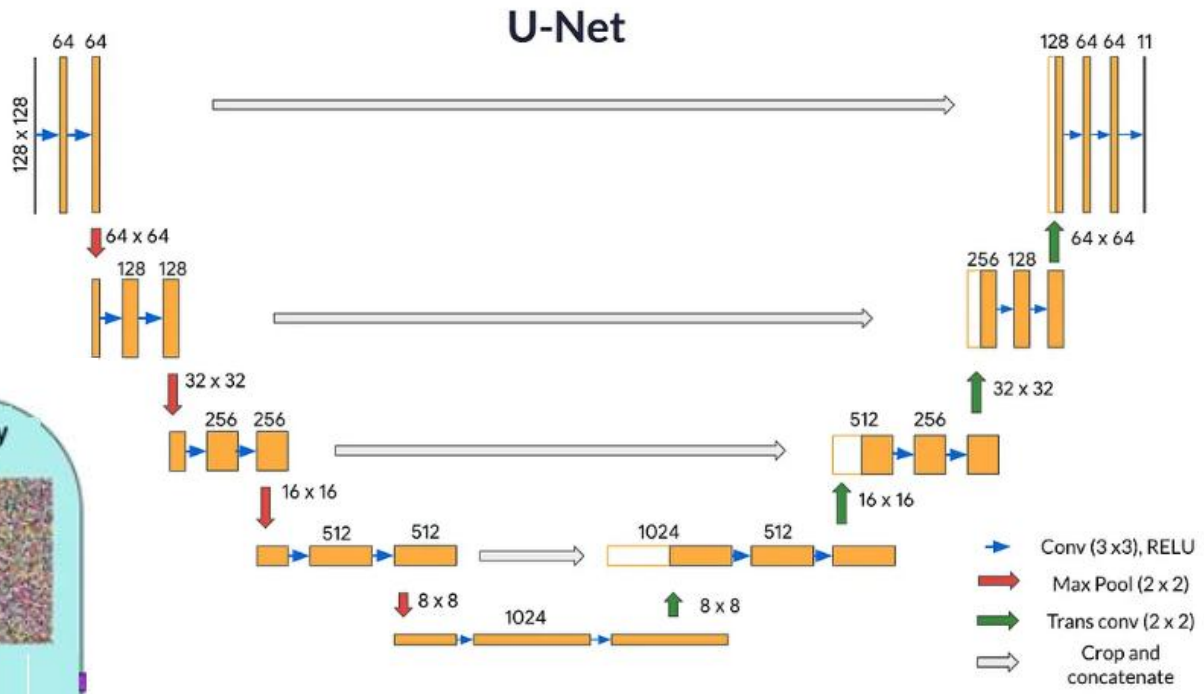
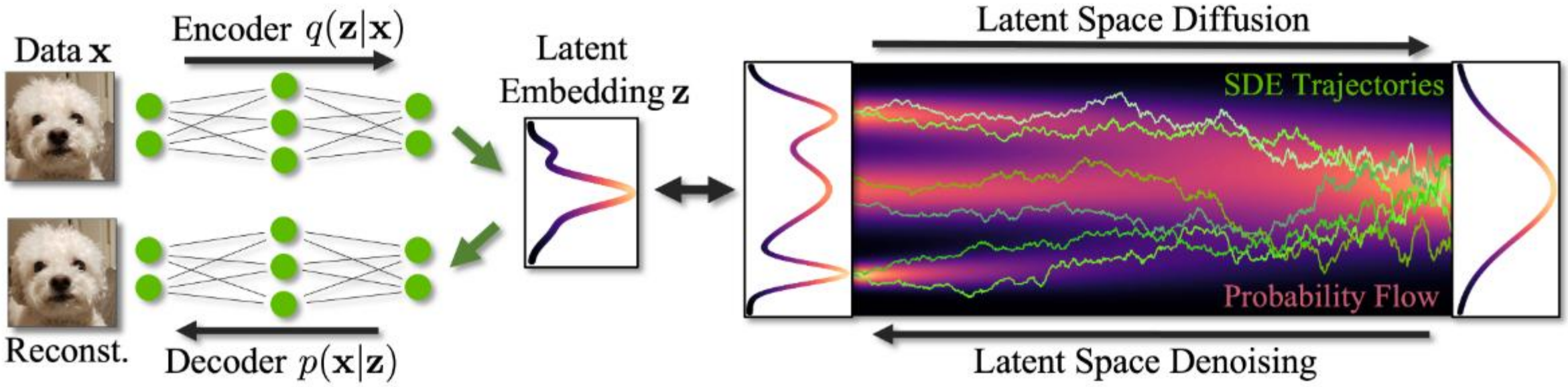




The U-Net architecture

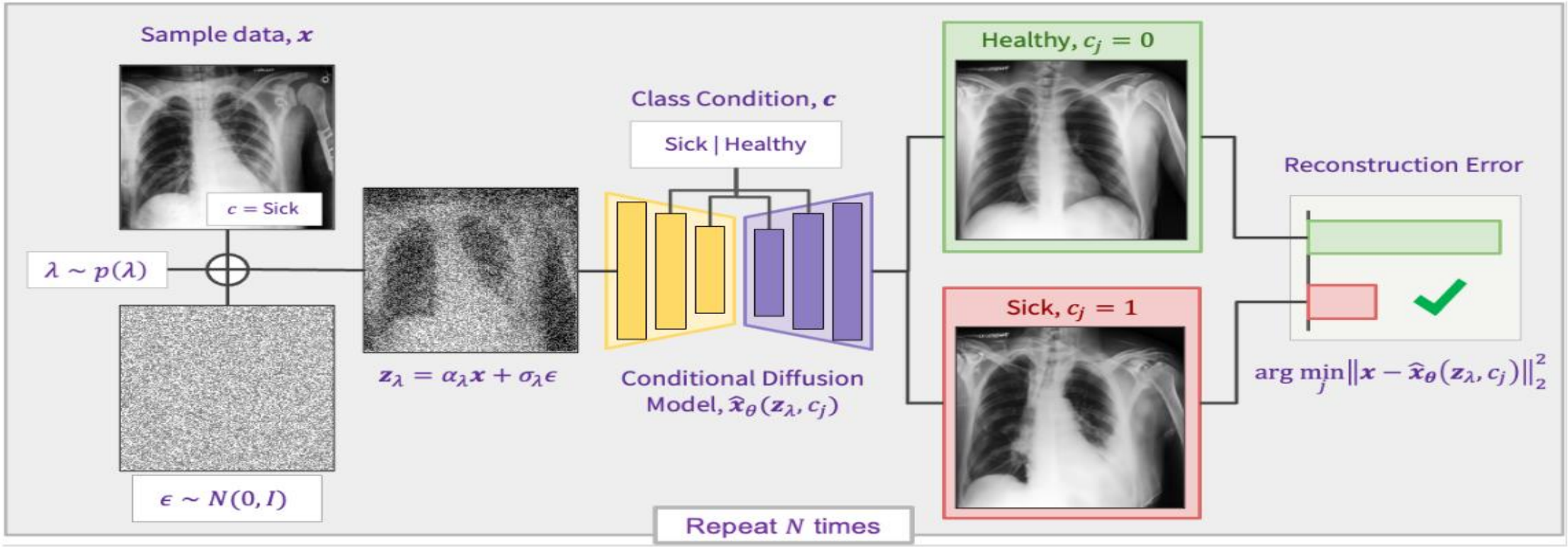
Latent Diffusion Models



- Operates on Latent Space
- Uses Autoencoders
- Increased Efficiency
- Reduces memory and computation cost
- Still produces high-resolution, high-quality images
- Used in Stable Diffusion, DALL·E 2, etc.

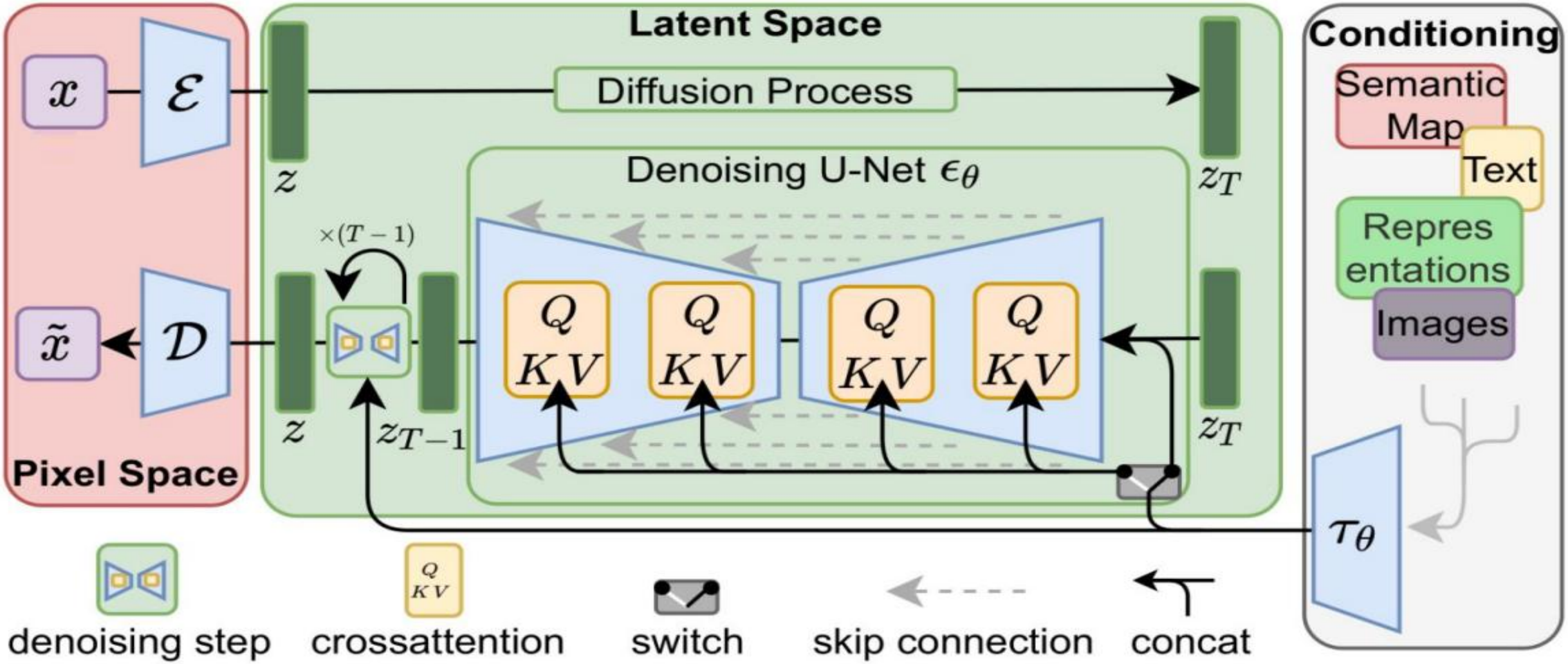
NeurIPS 2023 Tutorial: *Latent Diffusion Models: Is the Generative AI Revolution Happening in Latent Space?*

Conditioned Diffusion: Guiding image generation



- Conditioning inputs: text, image, or class labels
- Text encoder → embeddings
- Embeddings guide denoising at each step
- Enables controllable generation

Stable Diffusion Model



Diffusion Models

- **Top Stable Diffusion Models**

DALL·E 2 — photorealistic generation
Realism and semantic understanding

Stable Diffusion — open-source ecosystem
Can run locally

Midjourney — artistic image quality
Excels in art and aesthetics

Imagen — strong language alignment
Scaling language models can improve the quality

- **Top Applications**

Text-to-Image generation
Image editing & inpainting
Design & creative workflows
Medical imaging enhancement
Video & 3D generation

Thank You